

Flows in inkjet-printed aqueous rivulets

Vadim Bromberg, Timothy J. Singler

Mechanical Engineering Department,
State University of New York at Binghamton, Binghamton, NY 13902, USA

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Abstract

We used optical microscopy to investigate flows inside water rivulets that were inkjet-printed onto different surfaces and under different ambient conditions. The acquired fluid dynamics videos were submitted to the 2013 Gallery of Fluid Motion.

1 Introduction

Understanding the flow inside sessile liquid structures of different shapes is important in a variety of solution-based material deposition and patterning processes. Solvent evaporation inherent in these processes is already known to lead to a rich variety of flows [1]. The small length scale and general lack of shape symmetry implies the potential for capillary pressure gradients and corresponding flow phenomena. Finally, the non-instantaneous nature of the formation of these liquid structures adds another element to the flow complexity.

In the linked videos, we investigated the internal flow of inkjet-printed water rivulets of finite length using optical microscopy. Six millimeter-long rivulets were formed by printing a pre-determined number of drops ($D_0 = 52\mu\text{m}$) at controlled frequency ($f = 20\text{Hz}$) and spatial overlap ($\Delta x = 20\mu\text{m}$). Microscope cover slips made of borosilicate glass were surface coated and used as substrates. Two surface coatings were investigated - S1805 photoresist with and without KOH etching. Rivulets were printed inside a controlled humidity chamber at two relative humidity levels - 45% and 25%. The ambient, ink, and substrate were kept at room temperature ($T = 25\text{C}$). For fluorescent microscopy videos, the water was seeded

with Nile Red polystyrene spheres ($1.1\mu\text{m}$) at 0.1% volume fraction. We report the experimental details and results of a wider range of both printing parameters, ink properties, and substrate surfaces elsewhere [2].

The two surface coatings resulted in different values of the static advancing and receding contact angles (θ_A and θ_R) for water and in the rivulet formation process during printing. Water droplets on the photoresist-coated surface exhibited $\theta_A = 80^\circ$ and $\theta_R = 35^\circ$ while those on the etched surface $\theta_A = 45^\circ$ and $\theta_R = 0$. Complete rivulet formation was inhibited on the first surface type due to a non-zero θ_R , which allowed capillary-driven contact line recession and rivulet breakup into individual droplets [3]. The zero θ_R on the second surface type prevented rivulet breakup [3] but resulted in the formation of a distinctive bulge at the starting end of the water rivulet. The bulge grows immediately after the coalescence of the first few drops during rivulet formation.

We investigated the flow that causes bulge growth by using optical fluorescent microscopy. During rivulet formation, a distinctly pulsatile axial flow drives fluid away from the terminal end of the growing rivulet where printed drops are landing. The frequency of the flow is the same as the drop frequency. It has been hypothesized that the large local mean curvature in the region where drops land causes a capillary pressure gradient along the length of the rivulet [4]. Using μPIV , the height- and width-averaged axial velocity was measured and showed that the amplitude of the pulsatile flow decreases with reduced relative humidity in the ambient, all other conditions being fixed. The decreased flow magnitude results in the inhibition of bulge formation.

2 Videos

The video entry to the 2013 Gallery of Fluid motion is shown in

- Video 1 - Low resolution
- Video 2 - High resolution

In the video, the following subjects are shown:

- A schematic of the flow observation approach
- Visualization of rivulet break up during printing onto a substrate with non-zero receding contact angle (surface type 1)

- Visualization of successful rivulet formation on surface type 2 and consequent bulge growth
- Fluorescence microscopy video showing pulsatile axial flow and corresponding mean axial speed
- The effect of liquid evaporation on axial flow and bulge growth inhibition

References

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